

DATASCI 350 - Data Science Computing

Assignment 01 - Computational Literacy

Instructions

This assignment evaluates your understanding of topics covered in the first two weeks of class, including binary and hexadecimal number systems, ASCII encoding, and programming language fundamentals.

You must complete this assignment individually. You may use notes, books, and AI tools, but you must submit original work. Any errors introduced by AI tools are your responsibility, so always double-check your answers. Acknowledge all resources used, including input from classmates and AI, in a short note at the end of your submission. If you are unsure about permissible resources or proper acknowledgement, please ask the instructor.

Present your solutions clearly and systematically, showing your problem-solving process. For every conversion question, show your steps: an answer without working receives partial credit only.

Submission

Please submit your solutions as either a single Jupyter notebook or a PDF file. Answer each question in the empty cell below it. Submit your completed assignment to Canvas.

Question 01

Convert the decimal number 77 to binary. Show your work.

1001101, because $(1 \times 2^0) + (0 \times 2^1) + (1 \times 2^2) + (1 \times 2^3) + (0 \times 2^4) + (0 \times 2^5) + (1 \times 2^6) = 1 + 4 + 8 + 64 = 77$

Question 02

Convert the binary number 10110110 to decimal. Show your work.

182, because $(0 \times 2^0) + (1 \times 2^1) + (1 \times 2^2) + (0 \times 2^3) + (1 \times 2^4) + (1 \times 2^5) + (0 \times 2^6) + (1 \times 2^7) = 2 + 4 + 16 + 32 + 128 = 182$

Question 03

Convert the decimal number 2026 to hexadecimal. Show the division steps.

7EA, because $2026 / 16 = 126$ (Remainder: 10 = A), $126 / 16 = 7$ (Remainder: 14 = E), $7 / 16 = 0$ (Remainder: 7 = 7)

Question 04

A designer gives you the colour RGB (46, 139, 87). Write it as a hexadecimal colour code (the six-digit form used on the web, starting with #). Show how you convert each channel.

#2E8B57 , because

$46 / 16 = 2$ (Remainder: 14 = E), $2 / 16 = 0$ (Remainder: 2 = 2)

$139 / 16 = 8$ (Remainder: 11 = B), $8 / 16 = 0$ (Remainder: 8 = 8)

$87 / 16 = 5$ (Remainder: 7 = 7), $5 / 16 = 0$ (Remainder: 5 = 5)

Question 05

A website uses the background colour **#7B2D26** . Convert it to its RGB components. Show your steps.

RGB (123, 45, 38), because

$7B = 7 \times 16 + B = 112 + 11 = 123$

$2D = 2 \times 16 + D = 32 + 13 = 45$

$26 = 2 \times 16 + 6 = 32 + 6 = 38$

Question 06

In two or three sentences: why do programmers write bytes in hexadecimal rather than binary? Use the fact that one hexadecimal digit represents exactly four bits.

Programmers write bytes in hexadecimal rather than binary because it is much more compact. A byte can be represented with just 2 hexadecimal digits, rather than 8 binary digits because a hexadecimal digit corresponds to 4 binary bits and 8 binary bits are one byte.

Question 07

Decode the following binary ASCII message. Each group of eight bits is one character.

01000100 01010011

Hint: ASCII codes for the uppercase letters run consecutively from A = 65 to Z = 90.

Show how you convert each group.

DS, because

$$01000100 = (0 \times 2^0) + (0 \times 2^1) + (1 \times 2^2) + (0 \times 2^3) + (0 \times 2^4) + (0 \times 2^5) + (1 \times 2^6) + (0 \times 2^7) = 68 = D$$

$$01010011 = (1 \times 2^0) + (1 \times 2^1) + (0 \times 2^2) + (0 \times 2^3) + (1 \times 2^4) + (0 \times 2^5) + (1 \times 2^6) + (0 \times 2^7) = 83 = S$$

Question 08

A byte has 8 bits. How many different values can one byte represent, and what is the largest number it can store? Explain briefly why the largest number is not 256.

Since a byte has 8 bits, 1 byte can represent $2^8 = 256$ different values. The largest number a byte can store would correspond to $11111111 = (1 \times 2^0) + (1 \times 2^1) + (1 \times 2^2) + (1 \times 2^3) + (1 \times 2^4) + (1 \times 2^5) + (1 \times 2^6) + (1 \times 2^7) = 255$. The largest number is not 256 because 0 counts as one of the 256 values a byte can represent, meaning that, in general, n bits can reach up to $2^n - 1$. So, 8 bits (1 byte) can reach up to $2^8 - 1 = 255$.

Question 09

Unicode contains over a million characters, yet a plain-English text file takes exactly the same space in UTF-8 as it would in ASCII. Explain why, in two or three sentences.

A plain-English text file takes exactly the same space in UTF-8 as it would in ASCII because although unicode contains over 1 million characters, its first 128 characters are ASCII. Since plain-English would contain these 128 characters, each character would correspond to 1 byte in both UTF-8 and ASCII.

Question 10

Give two differences between an interpreted and a compiled programming language, and one example of each.

1.) Interpreted programming languages interpret and execute commands immediately by running inside a program, while compiled languages convert code into binary instructions prior to executing it.

2.) Compiled languages (like C and C++) are generally considered 'closer to the machine', meaning the user has more control over computer resources like RAM and CPU usage, while interpreted languages usually provide less control over these.

C++ is an example of a compiled programming language.

R is an example of an interpreted programming language.

Question 11

Describe the Von Neumann bottleneck in two sentences, and name one hardware feature that reduces its impact.

The Von Neumann bottleneck occurs because the wires connecting the CPU to memory can only move either instructions or data at once since they carry both. This results in the CPU waiting for memory, causing it to be inactive while it waits. Caches are a hardware feature that reduces its impact.

Question 12

Verify three of your earlier answers with Python. In the code cell below, complete the three function calls and run the cell:

- use `bin()` to check your answer to Question 01,
- use `hex()` to check your answer to Question 03,
- use `chr()` to check your answer to Question 07.

Your manual working in the earlier questions is still required; this question only confirms it. In one sentence, note anything Python prints differently from your hand-written answer (for example, the `0b` and `0x` prefixes).

Differences between my hand-written answers and what Python prints include the following: Python prints the prefix "0b" for binary numbers (Question 01) and "0x" for hexadecimal numbers (Question 03), and Python does not print the hexadecimal characters as capital letters, while I do (Question 03).

```
In [4]: # Complete the calls and run this cell
print(bin(77))    # Question 01: put the decimal number inside the call
print(hex(2026))  # Question 03: put the decimal number inside the call
print(chr(68))    # Question 07: one call per character code
print(chr(83))
```

0b1001101

0x7ea

D

S

Resources Used

I used the "Lecture 02: Computational Literacy" notes

(<https://danilofreire.github.io/datasci350/lectures/lecture-02/02-computational-literacy.html>) from class to complete this assignment.